Question 3

a)

For hue: because hue is the name of the color, and when we put the RGB values into color converter, or just do estimation from larger R and G values than B, we can get a result of yellow color, so that the hue is yellow.

For saturation: we know that the saturation describe the purity of the color, and a fully saturated color is one with no mixture of white, which means there should be at least one value in RGB is 0. However, they are all larger than 0, which means the color is unsaturated.

For brightness: it means the extent to which an area appears to emit light, and we can see if a color is bright, at least one value in RGB should be 255. However, the biggest value in RGB is 200, which means the brightness of this color is relatively low.

b)

i) to prevent aliasing, we should know exactly about the highest frequency of sound, and sample twice as often as the highest frequency.

ii) actually, the appearance of an image should be roughly the same after subsampling. Because during the process, some of the bits used to represent color information will be thrown away. But human eye is more sensitive to changes in light than in color, which means this change of subsampled is out of human's perception. Therefore, the appearance stays the same.

c)

i) true color: 24 bits.

Size = 24 bits * 200 * 300 pixels = 180000 bytes.

ii) grayscale color: 8 bits.

Size = 8 bits * 200 * 300 pixels = 60000 bytes.

d)

i) for voiced sounds, their spectrum shows characteristic maxima, and these maxima is formants, implying that energy in the region is relatively concentrated. Also, in spectrogram, the bar parallel to the horizontal axis is formants. Because different people have different ways of pronunciation, different vowels will also be pronounced differently, the timbre of the voice is then different, and this difference can be directly expressed through the formant. Therefore, by using the formant, different vowels can be discovered, so that sound can be transcribed into text, which is doing speech recognition.

ii) psychoacoustics is a branch of science studying the psychological and physiological responses associated with sound, including noise, speech, and music. It shows how the human ears and brain perceive sound. For example, it identifies the threshold of hearing, meaning that some sounds that may be too quiet or too loud to be heard. Also, it identifies sounds that may be masked by some other sounds, this sound doesn't affect perception of the sound signal.

Then, as for sound compression, it will throw away data that doesn't matter, which should use threshold of hearing to detect and judge, as well as discard masked sound. And this information can be got through psychoacoustics. Therefore, it helps to drop data which is out of perception, and achieve lossy compression of sound.

Question 4

a)

i) RLE will work best. Because large areas of uniform color show spatial redundancy, which will be exploited during RLE compression. And the process of RLE is to physically reduce any type of repeating sequence, once the sequence reach a predefined number of occurrence. So, instead of storing each pixel as an individual value for uniform color, RLE chooses to store number pairs (c, n), where c indicates the color and n indicates the length.

ii) JPEG is lossy, because it throws away information during process, like the fine details in high frequency in DCT domain, and the color information through sub-sampled.

But because human eye is more sensitive to changes in light than in color, the color information thrown during sub-sampling won't be captured by human eyes. Also, those fine details in high frequency components are beyond the acuity of human perception, that is, the abrupt change by high frequency component can't be seen. Therefore, some JPEG images look like the original image after decompression.

iii) because all B frames rely on two references -- previous and future I and P frames and make most prediction, so there will be only little change and very similar between B frames and the reference frames. Plus, B frames utilize inter-frame compression (temporal compression), which just encode the change from one frame to the next, so actually B frames just store small data and achieve high compression.